

Ex.9 Reader writer

Develop a c program to implement reader writer problem. Consider multiple readers and writers and simulate the scenario for a) writers with priority and b) readers with priority. Demonstrate output with multiple windows

AIM:

To develop a c program to implement reader writer problem.

CODE:

READER WRITER MAIN:

```
#include<stdio.h>
#include<stdlib.h>
#include<sys/ipc.h>
#include<sys/shm.h>
#include<string.h>
#include<unistd.h>
#define SHMSZ 1024
int main() {
    key_t key1, key2;
    int shmid;
    int sec_id;
    char *shm, *k, *s; // Data
    char *ssm, *j, *t;
    key2 = 3400; // Pattern
    key1 = 3415; // Data
    if ((shmid = shmget(key1, SHMSZ, IPC_CREAT | 0666)) < 0) {
        perror("shmget");
        exit(1);
    }
    if ((shm = shmat(shmid, NULL, 0)) == (char *) -1) {
        perror("shmat");
        exit(1);
    }
    //*****
    if ((sec_id = shmget(key2, SHMSZ, IPC_CREAT | 0666)) < 0) {
        perror("shmget");
        exit(1);
    }
}
```

```

if ((ssm = shmat(sec_id, NULL, 0)) == (char *) -1) {
    perror("shmat");
    exit(1);
}
t = ssm;
s = shm;
char pattern[10];
printf("\nEnter the pattern in caps: ");
fgets(pattern, 10, stdin); // Will accept till ENTER key is hit...
int len = strlen(pattern); // Corrected length calculation
for (int i = 0; i < len; i++) {
    *t = pattern[i];
    *s = pattern[i];
    t++;
    s++;
}
// Print the pattern written to shared memory
printf("\nPattern written to shm mem of main : %s\nPattern written to wwp & rwp shm mem: ", ssm);
for(char* i=shm; *i!='\0'; i++)
    printf("%c", *i);
putchar('\n');

int choice;
printf("1. READER PRIORITY\n2. WRITER PRIORITY\n3. BASED ON FIRST\n4. AS PER GIVEN\n");
printf("Enter the priority (1/2/3/4): ");
scanf("%d", &choice);
if (choice == 1)
{
    // Reader Priority
    char first = *ssm;
    printf("\nThe first character is: %c", first);
    int rcount = 0;
    int wcount = 0;
    for (k = ssm; *k != '\0'; k++) {
        if (*k == 'R')
            rcount++;
        else if (*k == 'W')
            wcount++;
    }
    printf("\nThe no. of readers is: %d", rcount);
    printf("\nThe no. of writers is: %d\n", wcount);
}

```

```

for (int i = 0; i < rcount; i++) {
    system("gnome-terminal -- ./r.out");
    sleep(5);
}
for (int i = 0; i < wcount; i++) {
    system("gnome-terminal -- ./w.out");
    sleep(10);
}
}
else if (choice == 2)
{
    // Writer Priority
    char first = *ssm;
    printf("\nThe first character is: %c", first);
    int rcount = 0;
    int wcount = 0;
    for (k = ssm; *k != '\0'; k++) {
        if (*k == 'R')
            rcount++;
        else if (*k == 'W')
            wcount++;
    }
    printf("\nThe no. of readers is: %d", rcount);
    printf("\nThe no. of writers is: %d\n", wcount);

    // Open writer and reader processes in separate terminals
    for (int i = 0; i < wcount; i++) {
        system("gnome-terminal -- ./w.out");
        sleep(10);
    }
    for (int i = 0; i < rcount; i++) {
        system("gnome-terminal -- ./r.out");
        sleep(5);
    }
}
else if (choice == 3)
{
    // based on first
    char first = *ssm;
    printf("\nThe first character is: %c", first);
    char *h;
    h = ssm;
    if (*h == 'R') {
        int rcount = 0;

```

```

int wcount = 0;
for (k = h; *k != '\0'; k++) {
    if (*k == 'R')
        rcount++;
    else if (*k == 'W')
        wcount++;
} printf("\nThe no. of readers is: %d", rcount);
printf("\nThe no. of writers is: %d\n", wcount);
sleep(10);
for (int i = 0; i < rcount; i++) {
    system("gnome-terminal -- ./r.out");
    sleep(5);
}
for (int i = 0; i < wcount; i++) {
    system("gnome-terminal -- ./w.out");
    sleep(10);
}
} else if (*h == 'W') {
    int rcount = 0;
    int wcount = 0;
    for (k = h; *k != '\0'; k++) {
        if (*k == 'R')
            rcount++;
        else if (*k == 'W')
            wcount++;
    } printf("\nThe no. of readers is: %d", rcount);
    printf("\nThe no. of writers is: %d\n", wcount);
    sleep(5);
    for (int i = 0; i < wcount; i++) {
        system("gnome-terminal -- ./w.out");
        sleep(10);
    }
    for (int i = 0; i < rcount; i++) {
        system("gnome-terminal -- ./r.out");
        sleep(5);
    }
}
}
}
else if ( choice == 4 )
{
    // as per given
    char f=*ssm;
    char *h=ssm;

```

```

int rcount = 0;
int wcount = 0;
for (k = h; *k != '\0'; k++) {
    if (*k == 'R')
        rcount++;
    else if (*k == 'W')
        wcount++;
}
printf("\nThe first character is : %c",f);
printf("\nThe no. of readers is: %d", rcount);
printf("\nThe no. of writers is: %d\n", wcount);
while(*h != '\0')
{
    if(*h == 'R')
    {
        system("gnome-terminal -- ./r.out");
        sleep(5);
    }
    else if(*h == 'W')
    {
        system("gnome-terminal -- ./w.out");
        sleep(10);
    }
    h++;
}
else {
    printf("Invalid choice.\n");
}
// Detach and remove shared memory segments
shmdt(shm);
shmdt(ssm);
shmctl(shmid, IPC_RMID, NULL);
shmctl(sec_id, IPC_RMID, NULL);
return 0;
}

```

READER:

```

#include<stdio.h>
#include<stdlib.h>
#include<sys/types.h>
#include<sys/wait.h>
#include<unistd.h>

```

```

#include <sys/shm.h>
#define SHMSZ 1024
int main() {
    int shmid;
    key_t key;
    key = 3415;
    char *shm, *s, *k;
    if ((shmid = shmget(key, SHMSZ, 0666)) < 0) {
        perror("shmget");
        exit(1);
    }
    if ((shm = shmat(shmid, NULL, 0)) == (char *) -1) {
        perror("shmat");
        exit(1);
    }
    char flag = *shm;
    s = shm;
    printf("\nThe data at the first location of the Shared Memory is:%c\n", flag);
    printf("The values in the shared memory : %s\n",shm);
    // for (k = shm; *k != '\0'; k++) {
    // printf("%c", *k);
    // }
    putchar('\n');
    sleep(5);
    printf("\nThe Read process of the Reader is complete!!!\n");
    return 0;
}

```

WRITER:

```

#include<stdio.h>
#include<stdlib.h>
#include<sys/types.h>
#include<sys/wait.h>
#include<unistd.h>
#include <sys/shm.h>
#include<stdbool.h>
#define SHMSZ 1024
#define MAX_LIMIT 1024

int main() {
    int shmid;
    key_t key;
    key = 3415;

```

```

char *shm, *s, *k;
if ((shmid = shmget(key, SHMSZ, 0666)) < 0) {
    perror("shmget");
    exit(1);
}
if ((shm = shmat(shmid, NULL, 0)) == (char *) -1) {
    perror("shmat");
    exit(1);
}
s = shm;
printf("\n %d is the Write process ID that is writing into Shared Memory...\n", getpid());
while (*s != '\0')
    s++;
char x = *s; // The data here should be null
printf("\n The Data here should be null ( coz this is the end of shm) : %c\n", x);
char str[MAX_LIMIT];
int choice;
do {
    printf("\n Enter the String to be entered into the Shared Memory: ");
    fgets(str, MAX_LIMIT, stdin); // Will accept input till ENTER key is hit...
    printf("\nAre you sure the string you want to enter is: %s",str);
    printf("\nIf YES -> 1 | NO -> 0: ");
    scanf("%d", &choice);
    getchar(); // Consume the newline character left by scanf
} while (choice != 1);
printf("The String that will be entered into the SHM is: %s\n",str);
// Copy the string into shared memory
for (int i = 0; i < MAX_LIMIT; i++) {
    *s++ = str[i]; // After the last memory location, s will point to NULL
    if (str[i] == '\0')
        break;
}
sleep(5); // Simulate some processing time
printf("\nThe Write process for %d Process is complete!!!\n",getpid());
printf("Contents of the shared memory after writing: ");
for (k = shm; *k != '\0'; k++) {
    printf("%c", *k);
}
putchar('\n');
return 0;
}

```

SCREENSHOT:

READER PRIORITY:

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/shm.h>
#include <semaphore.h>
#include <string.h>

#define SHM_SIZE 1024
#define WRITER_SIZE 1024
#define READER_SIZE 1024

int main()
{
    int s;
    int key_t;
    key_t = 0;
    char *shm;
    if ((s = shm_open("/shm", O_CREAT, 0666)) < 0)
        perror("shm_open");
    if ((shm = mmap(0, SHM_SIZE, PROT_READ|PROT_WRITE, MAP_SHARED, s, 0)) < 0)
        perror("mmap");
    if ((key_t = shmctl(s, SHM_UNLINK, 0)) < 0)
        perror("shmctl");

    char *wshm;
    s = shm_open("/wshm", O_CREAT, 0666);
    printf("The data at the first location of the Shared Memory is:R\n");
    printf("The values in the shared memory : RWRW\n");

    // The Read process of the Reader is complete!!!\n");
    return 0;
}
```

```
guest-9cp7kb@admin25-VirtualBox: ~
guest-9cp7kb@admin25-VirtualBox:~$ gcc rw.c -o rw
guest-9cp7kb@admin25-VirtualBox:~$ gcc reader.c -o r.out
guest-9cp7kb@admin25-VirtualBox:~$ gcc writer.c -o w.out
guest-9cp7kb@admin25-VirtualBox:~$ ./rw

Enter the pattern in caps: RWRW

Pattern written to shm mem of main : RWRW

Pattern written to wwp & rwp shm mem: RWRW

1. READER PRIORITY
2. WRITER PRIORITY
3. BASED ON FIRST
4. AS PER GIVEN
Enter the priority (1/2/3/4): 1

The first character is: R
The no. of readers is: 2
The no. of writers is: 2
```

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/shm.h>
#include <semaphore.h>
#include <string.h>

#define SHM_SIZE 1024
#define WRITER_SIZE 1024
#define READER_SIZE 1024

int main()
{
    int s;
    int key_t;
    key_t = 0;
    char *shm;
    if ((s = shm_open("/shm", O_CREAT, 0666)) < 0)
        perror("shm_open");
    if ((shm = mmap(0, SHM_SIZE, PROT_READ|PROT_WRITE, MAP_SHARED, s, 0)) < 0)
        perror("mmap");
    if ((key_t = shmctl(s, SHM_UNLINK, 0)) < 0)
        perror("shmctl");

    char *wshm;
    s = shm_open("/wshm", O_CREAT, 0666);
    printf("The data at the first location of the Shared Memory is:R\n");
    printf("The values in the shared memory : RWRW\n");

    // The Read process of the Reader is complete!!!\n");
    return 0;
}
```

```
guest-9cp7kb@admin25-VirtualBox: ~
guest-9cp7kb@admin25-VirtualBox:~$ gcc rw.c -o rw
guest-9cp7kb@admin25-VirtualBox:~$ gcc reader.c -o r.out
guest-9cp7kb@admin25-VirtualBox:~$ gcc writer.c -o w.out
guest-9cp7kb@admin25-VirtualBox:~$ ./rw

Enter the pattern in caps: RWRW

Pattern written to shm mem of main : RWRW

Pattern written to wwp & rwp shm mem: RWRW

1. READER PRIORITY
2. WRITER PRIORITY
3. BASED ON FIRST
4. AS PER GIVEN
Enter the priority (1/2/3/4): 1

The first character is: R
The no. of readers is: 2
The no. of writers is: 2
```

```
Activities Terminal Thu 09:15
reader.c
Save
File Edit View Search Terminal Help
5043 is the Write process ID that is writing into Shared Memory...
The Data here should be null ( coz this is the end of shm ) :
Enter the String to be entered into the Shared Memory: HI
Are you sure the string you want to enter is: HI
If YES -> 1 | NO -> 0: 1
The Read process of the Reader is complete!!!\n");
return 0;
}

guest-9cp7kb@admin25-VirtualBox: ~
File Edit View Search Terminal Help
guest-9cp7kb@admin25-VirtualBox:~$ gcc rw.c -o rw
guest-9cp7kb@admin25-VirtualBox:~$ gcc reader.c -o r.out
guest-9cp7kb@admin25-VirtualBox:~$ gcc writer.c -o w.out
guest-9cp7kb@admin25-VirtualBox:~$ ./rw
Enter the pattern in caps: RWRW
Pattern written to shm mem of main : RWRW
Pattern written to wwp & rwp shm mem: RWRW
1. READER PRIORITY
2. WRITER PRIORITY
3. BASED ON FIRST
4. AS PER GIVEN
Enter the priority (1/2/3/4): 1
The first character is: R
The no. of readers is: 2
The no. of writers is: 2
```

```
Activities Terminal Thu 09:15
reader.c
Save
File Edit View Search Terminal Help
5053 is the Write process ID that is writing into Shared Memory...
The Data here should be null ( coz this is the end of shm ) :
Enter the String to be entered into the Shared Memory: HELLO
Are you sure the string you want to enter is: HELLO
If YES -> 1 | NO -> 0: 1
The String that will be entered into the SHM is: HELLO
char flag;
s = shm;
printf("\n");
printf("The Read process of the Reader is complete!!!\n");
return 0;
}

guest-9cp7kb@admin25-VirtualBox: ~
File Edit View Search Terminal Help
guest-9cp7kb@admin25-VirtualBox:~$ gcc rw.c -o rw
guest-9cp7kb@admin25-VirtualBox:~$ gcc reader.c -o r.out
guest-9cp7kb@admin25-VirtualBox:~$ gcc writer.c -o w.out
guest-9cp7kb@admin25-VirtualBox:~$ ./rw
Enter the pattern in caps: RWRW
Pattern written to shm mem of main : RWRW
Pattern written to wwp & rwp shm mem: RWRW
1. READER PRIORITY
2. WRITER PRIORITY
3. BASED ON FIRST
4. AS PER GIVEN
Enter the priority (1/2/3/4): 1
The first character is: R
The no. of readers is: 2
The no. of writers is: 2
```

WRITER PRIORITY:

The screenshot shows a terminal window with the following output:

```
#include <stdio.h>
#include <unistd.h>
#include <sys/shm.h>
#include <sys/types.h>
#include <stdlib.h>
#define SHM_SIZE 1024
int main()
{
    key_t key = 1234;
    char *shm;
    if (shmctl(key, SHM_RDONLY, 0666) < 0)
    {
        perror("shmctl");
        exit(1);
    }
    shm = shmat(key, 0, 0);
    if (shm == (char *)0)
    {
        perror("shmat");
        exit(1);
    }
    printf("5702 is the Write process ID that is writing into Shared Memory...\n");
    printf("The Data here should be null ( coz this is the end of shm ) : \n");
    Enter the String to be entered into the Shared Memory: HI
    Are you sure the string you want to enter is: HI
    If YES -> 1 | NO -> 0: 1
    The String that will be entered into the SHM is: HI
    printf("Contents of the shared memory after writing: ");
    for (k = shm; *k != '\0'; k++) {
        printf("%c", *k);
    }
    putchar('\n');
    return 0;
}
```

The output of the program is:

```
5702 is the Write process ID that is writing into Shared Memory...
The Data here should be null ( coz this is the end of shm ) :
Enter the String to be entered into the Shared Memory: HI
Are you sure the string you want to enter is: HI
If YES -> 1 | NO -> 0: 1
The String that will be entered into the SHM is: HI
Contents of the shared memory after writing: HI
```

The screenshot shows a terminal window with the following output:

```
#include <stdio.h>
#include <unistd.h>
#include <sys/shm.h>
#include <sys/types.h>
#include <stdlib.h>
#define SHM_SIZE 1024
int main()
{
    key_t key = 1234;
    char *shm;
    if (shmctl(key, SHM_RDONLY, 0666) < 0)
    {
        perror("shmctl");
        exit(1);
    }
    shm = shmat(key, 0, 0);
    if (shm == (char *)0)
    {
        perror("shmat");
        exit(1);
    }
    printf("5715 is the Write process ID that is writing into Shared Memory...\n");
    printf("The Data here should be null ( coz this is the end of shm ) : \n");
    Enter the String to be entered into the Shared Memory: HELLO
    Are you sure the string you want to enter is: HELLO
    If YES -> 1 | NO -> 0: 1
    The String that will be entered into the SHM is: HELLO
    printf("Contents of the shared memory after writing: ");
    for (k = shm; *k != '\0'; k++) {
        printf("%c", *k);
    }
    putchar('\n');
    return 0;
}
```

The output of the program is:

```
5715 is the Write process ID that is writing into Shared Memory...
The Data here should be null ( coz this is the end of shm ) :
Enter the String to be entered into the Shared Memory: HELLO
Are you sure the string you want to enter is: HELLO
If YES -> 1 | NO -> 0: 1
The String that will be entered into the SHM is: HELLO
Contents of the shared memory after writing: HELLO
```

```
Activities Terminal Thu 09:28
writer.c
Save
Terminal
File Edit View Search Terminal Help
The data at the first location of the Shared Memory is:R
The values in the shared memory : RWRW
HI
HELLO
int s;
int s;
key_t
key_t
char
if ((
perro
exit(
)
if ((
perro
exit(
)
char
s = s;
printf
printf
// fo
// pr
// )
putchar
sleep(5); // Simulate some processing time
printf("\nthe Write process for %d Process is complete!!\n",getpid());
printf("\nContents of the shared memory after writing: ");
return 0;
for (k = shm; *k != '\0'; k++) {
printf("%c", *k);
}
putchar('\n');
return 0;
}

n", getpid());

n", x);

guest-9cp7kb@admin25-VirtualBox: ~
File Edit View Search Terminal Help
guest-9cp7kb@admin25-VirtualBox:~$ gcc rw.c -o rw
guest-9cp7kb@admin25-VirtualBox:~$ gcc reader.c -o r.out
guest-9cp7kb@admin25-VirtualBox:~$ gcc writer.c -o w.out
guest-9cp7kb@admin25-VirtualBox:~$ ./rw

Enter the pattern in caps: RWRW

Pattern written to shm mem of main : RWRW

Pattern written to wwp & rwp shm mem: RWRW

1. READER PRIORITY
2. WRITER PRIORITY
3. BASED ON FIRST
4. AS PER GIVEN
Enter the priority (1/2/3/4): 2

The first character is: R
The no. of readers is: 2
The no. of writers is: 2
C Tab Width: 8 Ln 38, Col 49 INS
```

```
Activities Terminal Thu 09:28
writer.c
Save
Terminal
File Edit View Search Terminal Help
The data at the first location of the Shared Memory is:R
The values in the shared memory : RWRW
HI
HELLO
int s;
int s;
key_t
key_t
char
if ((
perro
exit(
)
if ((
perro
exit(
)
char
s = s;
printf
printf
// fo
// pr
// )
putchar
sleep(5); // Simulate some processing time
printf("\nthe Write process for %d Process is complete!!\n",getpid());
printf("\nContents of the shared memory after writing: ");
return 0;
for (k = shm; *k != '\0'; k++) {
printf("%c", *k);
}
putchar('\n');
return 0;
}

n", getpid());

n", x);

guest-9cp7kb@admin25-VirtualBox: ~
File Edit View Search Terminal Help
guest-9cp7kb@admin25-VirtualBox:~$ gcc rw.c -o rw
guest-9cp7kb@admin25-VirtualBox:~$ gcc reader.c -o r.out
guest-9cp7kb@admin25-VirtualBox:~$ gcc writer.c -o w.out
guest-9cp7kb@admin25-VirtualBox:~$ ./rw

Enter the pattern in caps: RWRW

Pattern written to shm mem of main : RWRW

Pattern written to wwp & rwp shm mem: RWRW

1. READER PRIORITY
2. WRITER PRIORITY
3. BASED ON FIRST
4. AS PER GIVEN
Enter the priority (1/2/3/4): 2

The first character is: R
The no. of readers is: 2
The no. of writers is: 2
C Tab Width: 8 Ln 38, Col 49 INS
```

FCFS:

The screenshot shows a Linux desktop with three terminal windows. The leftmost terminal window displays the C code for a shared memory program. The middle terminal window shows the execution of the program, including prompts for pattern, priority, and readers/writers. The rightmost terminal window shows the execution of the program on a virtual machine, also including prompts for pattern, priority, and readers/writers.

```
Activities Terminal Thu 09:38
writer.c Save
Terminal
File Edit View Search Terminal Help
6397 is the Write process ID that is writing into Shared Memory...
The Data here should be null ( coz this is the end of shm ) :
Enter the String to be entered into the Shared Memory: WELCOME
Are you sure the string you want to enter is: WELCOME
If YES -> 1 | NO -> 0: 1
The String that will be entered into the SHM is: WELCOME
// for priority
// pr
// )
putchar('\n');
sleep(5); // Simulate some processing time
printf("\n\nThe Write process for %d Process is complete!!!\n",getpid());
printf("\n\nContents of the shared memory after writing: ");
return 0;
for (k = shm; *k != '\0'; k++) {
printf("%c", *k);
}
putchar('\n');
return 0;
}

guest-9cp7kb@admin25-VirtualBox: ~
File Edit View Search Terminal Help
guest-9cp7kb@admin25-VirtualBox:~$ ./rw
Enter the pattern in caps: WRWR
Pattern written to shm mem of main : WRWR
Pattern written to wwp & rwp shm mem: WRWR
1. READER PRIORITY
2. WRITER PRIORITY
3. BASED ON FIRST
4. AS PER GIVEN
Enter the priority (1/2/3/4): 4
The first character is : W
The no. of readers is: 2
The no. of writers is: 2
guest-9cp7kb@admin25-VirtualBox:~$ ./rw
Enter the pattern in caps: WRWR
Pattern written to shm mem of main : WRWR
Pattern written to wwp & rwp shm mem: WRWR
1. READER PRIORITY
2. WRITER PRIORITY
3. BASED ON FIRST
4. AS PER GIVEN
Enter the priority (1/2/3/4): 4
The first character is : W
The no. of readers is: 2
The no. of writers is: 2
```

```
Activities Terminal Thu 09:38
writer.c Save
Terminal
File Edit View Search Terminal Help
The data at the first location of the Shared Memory is:W
The values in the shared memory : WRWR
HI
int s
int s
key_t
key_t
char
char
if ((
perro
exit(
)
if ((
perro
exit(
)
char
s = s
print
print
// fo
// pr
// )
putchar('\n');
sleep(5); // Simulate some processing time
printf("\n\nThe Write process for %d Process is complete!!!\n",getpid());
printf("\n\nContents of the shared memory after writing: ");
return 0;
for (k = shm; *k != '\0'; k++) {
printf("%c", *k);
}
putchar('\n');
return 0;
}

guest-9cp7kb@admin25-VirtualBox: ~
File Edit View Search Terminal Help
guest-9cp7kb@admin25-VirtualBox:~$ ./rw
Enter the pattern in caps: WRWR
Pattern written to shm mem of main : WRWR
Pattern written to wwp & rwp shm mem: WRWR
1. READER PRIORITY
2. WRITER PRIORITY
3. BASED ON FIRST
4. AS PER GIVEN
Enter the priority (1/2/3/4): 4
The first character is : W
The no. of readers is: 2
The no. of writers is: 2
guest-9cp7kb@admin25-VirtualBox:~$ ./rw
Enter the pattern in caps: WRWR
Pattern written to shm mem of main : WRWR
Pattern written to wwp & rwp shm mem: WRWR
1. READER PRIORITY
2. WRITER PRIORITY
3. BASED ON FIRST
4. AS PER GIVEN
Enter the priority (1/2/3/4): 4
The first character is : W
The no. of readers is: 2
The no. of writers is: 2
```